SCIBENCH ANONYMIZED REPLICATION DOSSIER
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Anonymous replication dossier: logarithmic spin effective Hamiltonian

Scope

Reproduce the deterministic core calculation that supplies an effective field to a finite-temperature spin-dynamics simulation. This task does not run the subsequent stochastic dynamics and does not claim to reproduce its sampled magnetization curves.

Consider one spin with dimensionless S_z eigenvalues m=-s,...,s and

[EQUATION]

```
    H = -(A_0 I + A_1 S_z + A_2 S_z^2),
    A_0 = lambda sigma,
    A_1 = g mu_B mu_0 H_z,
    A_2 = K-lambda sigma.
```

Use lambda sigma=0 and mu_0 H_z=1 tesla. Obtain the absolute electron Lande? factor, Bohr magneton, and Boltzmann constant from a standard CODATA constants library. Let beta=1/(k_B T).

Coherent-state construction

For p=0,...,2s, define |p>=|s,s-p> and

[EQUATION]

```
    |z> = (1+|z|^2)^(-s)
        sum_p binom(2s,p)^(1/2) z^p |p>.
```

Map the stereographic radius to the unit-spin orientation by

[EQUATION]

```
    u_z = (1-|z|^2)/(1+|z|^2),
    |z|^2 = (1-u_z)/(1+u_z).
```

The polynomial entering the coherent-state partition integrand is

[EQUATION]

```
    L = sum_p binom(2s,p) |z|^(2p)
        exp(beta A_2 p^2)
        exp[-beta(2s A_2+A_1)p].
```

After factoring out the energy of the m=s state, define

[EQUATION]

```
    W(beta,u_z) = L/(1+|z|^2)^(2s),
    H_eff(beta,u_z) = -beta^(-1) log W(beta,u_z).
```

This factorization fixes the otherwise arbitrary additive constant: H_eff(beta,1)=0. Submit the dimensionless gauge-fixed Hamiltonian

[EQUATION]

    h(beta,u_z) = H_eff(beta,u_z)/A_1.

Evaluate logarithms and exponential sums stably. In particular, do not form large low-temperature exponentials before applying the logarithm.

The longitudinal effective field supplied to spin dynamics is

[EQUATION]

    B_eff,z = -(g mu_B s)^(-1) partial H_eff/partial u_z.

Because A_1=g mu_B mu_0 H_z, submit the dimensionless field

[EQUATION]

    b(beta,u_z) = B_eff,z/(mu_0 H_z)
        = -(1/s) partial h/partial u_z.

The field should be obtained analytically or to equivalent numerical accuracy. It is not a thermal average and cannot be replaced by a partition-function magnetization.

Parameter groups and artifacts

Submit ‘temperature‘ as float64 ‘linspace(0.02,10.0,200)‘ in kelvin, shape ‘(200,)‘. Submit ‘orientation‘ as float64 ‘linspace(-0.98,0.98,201)‘, shape ‘(201,)‘. Array axes are always parameter row, increasing temperature, increasing orientation.

For the Figure 2 parameter group, use K/A_1=-2 and rows ordered by s={0.5,1.0,1.5,2.0}. Submit ‘figure2_hamiltonian‘ and ‘figure2_field‘, each with shape ‘(4,200,201)‘.

For the Figure 3 parameter group, fix s=1 and order rows by K/A_1={10,0,-1,-10}. Submit ‘figure3_hamiltonian‘ and ‘figure3_field‘, each with shape ‘(4,200,201)‘.

All six artifacts must be finite, numeric, non-pickle float64 NPY files. Graphics, stochastic trajectories, magnetization averages, and self-reported checkpoints are not scored.

[MASKED: all effective-Hamiltonian values, effective-field values, comparisons, and original graphics are withheld.]